In-class assignment: Sorting part of an array

COSC 2230 (CS III) Fall 2026 | Augustana U. | Friday, November 6, 2026

How this works

- On paper, 30 minutes. You may use your own handwritten notes. No printed slides, no laptop, no phone, no AI.
- Write your name at the top and answer below each task.
- **Write clearly.** If I cannot read your answer, it gets zero points.
- Syntax does not count: a missing semicolon or bracket costs nothing. An answer in plain words or pseudocode gets partial credit.
- 10 points: task 1: 5, task 2: 2, task 3: 3.

Name: _____

The code from class

```
public class Merge
{
    private static String[] aux;

    public static void merge(String[] a, int lo, int mid, int hi)
    {
        int i = lo, j = mid, N = hi - lo;
        for (int k = 0; k < N; k++)
        {
            if      (i == mid)                 aux[k] = a[j++];
            else if (j == hi)                  aux[k] = a[i++];
            else if (a[j].compareTo(a[i]) < 0) aux[k] = a[j++];
            else                               aux[k] = a[i++];
        }
        for (int k = 0; k < N; k++)
            a[lo + k] = aux[k];
    }

    public static void sort(String[] a)
    {
        aux = new String[a.length];
        sort(a, 0, a.length);
    }

    public static void sort(String[] a, int lo, int hi)
    {
        int N = hi - lo;
        if (N <= 1) return;
        int mid = lo + N/2;
        sort(a, lo, mid);
        sort(a, mid, hi);
        merge(a, lo, mid, hi);
    }
}
```

Task 1: Trace (5 points)

`Merge.sort(a)` sorts this array:

```
it was the best of times
```

After each call of `merge()`, write down `lo`, `mid`, `hi` and the whole array, in the order in which the merges happen. You may not need every row.

lo	mid	hi	a[0]	a[1]	a[2]	a[3]	a[4]	a[5]

Task 2: Find the bug (2 points)

Another program sorts only part of an array. It does not call `sort(a)`:

```
String[] a = { "it", "was", "the", "best", "of", "times" };
Merge.sort(a, 1, 5);
```

What happens when it runs, and why?

Task 3: Fix it (3 points)

Change `Merge` so that `sort(a, lo, hi)` also works when a program calls it directly. `sort(a)` must keep working, and `merge()` stays as it is. Fill in the three methods:

```
public static void sort(String[] a)
{


}

public static void sort(String[] a, int lo, int hi)
{


}

private static void sortRange(String[] a, int lo, int hi)
{







}
```

How big must `aux` be at least?